

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Annexure A

Total Marks:25

Comparative Analysis

Code of Conduct:

Shanika (192511387)

I Betty A. (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Betty

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

1) Star Topology vs Bus Topology

Cost:

Star Topology is more expensive because it requires a central switch / hub and individual cables.

Bus Topology: uses a single backbone cable, making it cheaper and easier to install.

Reliability:

Star is more reliable because failure of one node does not affect other nodes.

Traffic:

Bus topology can experience more collisions when many devices are connected.

Performance:

Star provides better performance because data transmission is controlled through the central device.

Troubleshooting:

Faults are easier to identify in star topology because each device has a separate connection.

Scalability:

Star can be expanded by connecting additional devices, depending on the switch capacity.

Conclusion:

Star is better for reliable and high-performance networks, while bus is suitable for small and low-cost networks.

2. Mesh Topology vs Star Topology:

Fault Tolerance:

Mesh topology has very high fault tolerance because it provides multiple paths.

Alternative paths:

If one link fails, data can travel through another available path.

Star Topology:

Has moderate fault tolerance because all devices depend on the central switch/hub.

Central Device Failure:

Failure of the central device can stop communication in the entire star network.

Reliability :

mesh is more reliable for critical network environment

cost :

mesh is generally more expensive because it requires many connections.

complexity :

mesh is more complex to install and manage than star.

conclusion :

mesh is preferred for critical applications where reliability is more important than cost and complexity.

3. Bus Topology vs mesh Topology

Installation:

Bus topology is simple and easy to install using a single backbone cable

Mesh topology:

Requires multiple connections between nodes, making installation complex.

Cost:

Bus is cheaper because it requires fewer cables and network connections.

Complexity:

Mesh becomes increasingly complex as more devices are added.

Maintenance:

Bus is easier to set up but backbone cable faults can affect the whole network.

Reliability:

Mesh provides higher reliability because alternative communication paths are available.

Scalability:

Adding many devices to a mesh network increases the number of required connections.

Conclusion:

Bus is suitable for small and simple networks, whereas mesh is suitable for networks requiring high reliability.

4. Hybrid Topology vs Star Topology

Scalability :

Hybrid topology is highly scalable because it combines two or more topologies

Flexibility :

Different network structures can be combined according to organizational requirements.

Expansion :

New network segments can be added without affecting the existing structure.

Star Topology :

Its expansion depends on the number of available ports on the central switch / hub.

Additional Hardware:

Star may require another switch when the existing device reaches its capacity.

Management:

Hybrid networks may require more careful administration because different topologies are combined.

Applications:

Hybrid topology provides useful for large organizations with different networking requirements.

Conclusion:

Hybrid topology provides greater flexibility and scalability than star topology.

5. Maintenance comparison of star, mesh, Bus and Hybrid

Star:

Easy to maintain because faults can be isolated to a particular node or cable.

Mesh:

Difficult to maintain because it has large number of connections.

Bus:

Backbone cable failure can affect the entire network.

Hybrid:

Has moderate maintenance complexity because multiple topologies are combined.

Troubleshooting:

Star makes troubleshooting easier because each device has an individual connection.

Mesh:

Troubleshooting can take more time due to its complex structure.

Administration:

Hybrid networks require careful monitoring and skilled administration.

Conclusion:

Star is generally the easiest to maintain, while mesh is more difficult and time-consuming.